



Grade 21-7-14
Sample # 240811

Method ReportCard
Lab Code 0584



Magruder Fertilizer Proficiency Testing Program
Regular PT Scheme

Proficiency For 4 Methods

Issue Date : 09/30/2024

Method Code	Analyte Name and Method (Units)	Lab 0584 Data		*Robust Method Values					Magruder Z Score	Method IA Status	Flag
		Value	range	Rob Mean	Rob SD	%RSD	Rob R-bar (n)	# Tests			
001.99	Ammoniacal N, Other (%)	11.08	0.1100	11.08	0.1324	1.19%	0.0772 (21)	22	-0.07	NA	0
002.99	Nitrate N, Other (%)	9.875	0.0700	9.903	0.3358	3.39%	0.1001 (21)	21	-0.08	NA	0
041.99	Direct Available P2O5, Other (7%)	7.025	0.0900	7.075	0.1366	1.93%	0.0817 (8)	8	-0.33	OK	0
148.01	Total S, Gravimetric - sulfate and elemental (5.1%)	5.095	0.0300	5.250	0.1742	3.32%	0.0870 (8)	9	-0.82	OK	0

Notes and Flags:

1: Interpreting Z Scores: Red indicates a normally distributed Z value >3 or <-3 (requires action), Orange = Z between 2 and 3 or -2 and -3 (warning) and a PASS at Green, where Z < 2 and Z >-2 (OK at 95%). The 9 Flag indicates a data problem - scores not calculated.

2: Flags indicate data usage: 0 = Used, 1 = rejected for duplicates too far apart, 2 = rejected as extreme outlier, 3 = removed from stats, 4 = rejected due to 0s submitted, 5 = LOD. The 9 Flag indicates a data problem - scores not calculated.

3: Method codes in light green indicate a guaranteed analyte. IA Status describes where your result is relative to the Robust Mean $\pm$ IA. Green is within the IA range and Red is Higher or Lower than the IA range about the Robust mean.

4: *Robust statistics not used if < 6 labs used in calculations, in this case Values and Z Scores are grayed out and should be interpreted cautiously. R-bar is robustified to provide a fair measure of the mean range. Identical duplicates not included in calculation of Rob R-bar. The (n) in Rob R-bar (n) = number of tests used to calculate Rob R-bar.